

BPhO Computational Challenge 2025

Reuben Okotie & Corinna Zhang

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1 Extension: Mie Scattering

1.1 Context & Introduction

The initial aim of this project was to create an abstraction of Rayleigh Scattering using the framework of geometric optics, visualising how white light refracts when interacting with a medium of water droplets. The primary goal was to demonstrate why shorter wavelength light appears to scatter more than longer wavelength light, providing visual insight into the question: "Why is the sky blue?"

Rayleigh scattering is the scattering of light as it interferes with particles much shorter than its wavelength in an elastic manner. This is a quantum phenomenon where the photons scattering probability depends on wavelength. A simple form of this relationship can be shown with:

$$I \propto \frac{1}{\lambda^4}$$

However, this simple relationship doesn't capture the full physics involved. The full expression of:

$$I_{\text{obs}}(R, \theta) = I_{\text{beam}} \frac{8\pi^4 d^6}{R^2 \lambda^4} \left(\frac{n^2 - 1}{n^2 + 2} \right)^2 (1 + \cos^2 \theta)$$

Where:

$I_{\text{obs}}(R, \theta)$ is the intensity of light from the perspective of an observer at a given distance (R) and angle (θ), however this aspect can be vectorised.

I_{beam} is the initial intensity of the light ray

d is the particle diameter (which must be much smaller than the wavelength (λ))

n is the bulk refractive index of the medium that the light ray is passing through. The refractive index represents how light propagates through a medium.

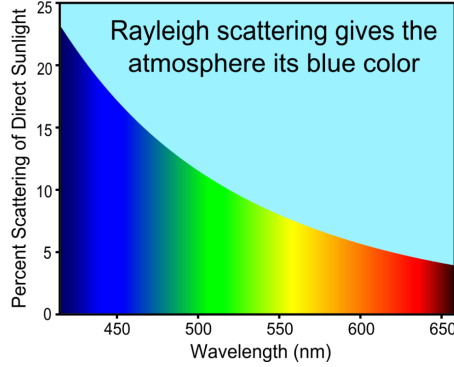


Figure 1: A clear model of how shorter wavelength scatters significantly more than red light. However, I was more interested to a model akin to an Agent-Based Model (using Object-Oriented Programming) to display this dynamically and hone the intuition to this. Source: Wikipedia

The objective was to demonstrate the effect of Rayleigh Scattering (the demonstration of the greater scattering of shorter wavelength light through a medium) through geometric optics, exploring whether it was possible to provide some basis for quantum phenomena through this topic.

Instead of implementing probabilistic scattering, I modelled a dense medium resembling a cloud to hopefully extract the same effect. Upon further investigation, Mie scattering proved more fitting for this approach, as it describes the interaction of light with particles comparable to or larger than the wavelength. Mie scattering explains why clouds and fog appear white despite containing the same water molecules that produce blue skies through Rayleigh scattering. While true Mie scattering requires solving Maxwell's equations, geometric optics provides a computationally efficient approximation in the large particle limit (particle diameter $\gg$ wavelength), which applies to typical cloud droplets (1-50 μm) interacting with visible light (0.4-0.7 μm).

This project was inspired by the desire to 'extract the essence' of a physical phenomenon and represent it in a visually compelling and interactive way. I find this necessary, especially for physics, as disparate areas tend to be related (often unified) in initially unintuitive ways. The model produced led to the discovery of a different, but equally interesting, physical system and is evidence of the necessity of these computational models.

1.2 Calculations

When trying to calculate the refraction of the ray using this equation, we settled on using vectors. Here is our thought process on calculating ray refraction.

Notation Used:

I is the vector of the incident ray

N is the vector of the normal to the surface of the raindrop. Therefore $\frac{N_y}{N_x}$ is the gradient of the normal

θ is the incident angle relative to the normal

ϕ is the angle of refraction

R is the vector for the refracted ray

n_1, n_2 are the refractive indices of the incident and refracted materials respectively

1.3 Refraction of Ray: Algebraic Approach

Using the dot product formula and rearranging, we got a formula for θ :

$$\vec{I} \cdot \vec{N} = |\vec{I}||\vec{N}| \cos \theta$$

$$\arccos \left(\frac{I_x N_x + I_y N_y}{|\vec{I}||\vec{N}|} \right) = \theta$$

Rearranging Snell's Law of Refraction ($n_1 \sin \theta = n_2 \sin \phi$) to get the dot product and its equivalence in terms of the refracted ray vector:

$$\phi = \arcsin \left(\frac{n_1 \sin \theta}{n_2} \right)$$

$$\vec{R} \cdot \vec{N} = |\vec{R}||\vec{N}| \cos \phi$$

Using Pythagoras to separate the x and y component of R , and rearrange to isolate R_x :

$$R_x^2 + R_y^2 = 1$$

$|\vec{R}| = 1$ has been set for ease of calculation, as we are only interested in the direction of the ray and not its magnitude. This normalised value has been used for the rest of the calculations in this section.

$$R_x = \sqrt{1 - R_y^2}$$

$$R_x N_x + R_y N_y = |\vec{N}| \cos \phi$$

Substitute R_x for into above equation and rearrange:

$$N_x \sqrt{1 - R_y^2} + R_y N_y = |\vec{N}| \cos \phi$$

$$\sqrt{1 - R_y^2} = \frac{|\vec{N}| \cos \phi - R_y N_y}{N_x}$$

Let $\gamma = |\vec{N}| \cos \phi$:

$$\frac{(\gamma - R_y N_y)^2}{N_x^2} = 1 - R_y$$

$$\left(\frac{N_y^2}{N_x^2}\right) R_y^2 + \left(\frac{1 - 2N_y \gamma}{N_x^2}\right) R_y + \left(\frac{\gamma^2}{N_x^2} - 1\right) = 0$$

Using these functions as a basis, a function was derived to calculate the two distinct y-values for the vector component of the refracted ray and its corresponding x-value.

Depending on whether the ray is entering/exiting the raindrop, the chosen vector will be the one with the minimum/maximum of the dot product between the incident and the refracted ray.

1.3.1 Reflection

On review, there was a simpler method that would have been derived - given my persistent use of the square root in calculations which only became apparent when testing variables in code.

1.4 Geometric Approach

The algebraic code often produced errors since floating-point inaccuracies were magnified during calculation. Therefore, a geometric approach was adopted that uses vector decomposition, which is both more stable and more computationally efficient. All vectors ($\vec{I}, \vec{N}, \vec{R}$) are assumed to be normalised (unit vectors).

The parallel component of the incident ray ($\vec{I}_{\parallel}$) is the fraction of the incident ray ($\vec{I}$) that is pointing in the direction of the normal ($\vec{N}$).

$$\vec{I}_{\parallel} = (\vec{I} \cdot \vec{N}) \vec{N}$$

Since $\vec{I} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$:

$$\vec{I}_{\perp} = 1 - \vec{I}_{\parallel}$$

Using Snell's Law of Refraction: $n_1 \sin \theta = n_2 \sin \phi$:

$$\vec{R}_{\perp} = \frac{n_1}{n_2} \vec{I}_{\perp}$$

$$\cos \phi = \vec{R}_{\parallel}$$

$$\cos \phi = \pm \sqrt{\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \left(\frac{n_1}{n_2} \vec{I}_{\perp}\right)}$$

Therefore:

$$\boxed{\vec{R} = \frac{n_1}{n_2} \vec{I}_{\perp} \pm \sqrt{\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \left(\frac{n_1}{n_2} \vec{I}_{\perp}\right)}}$$

Sign Convention: The negative will be chosen for entering rays, the positive for exiting rays. This is because the dot product is signed and indicates whether two vectors are pointing in the same direction.

This avoids quadratic solving entirely and gives the refracted ray directly.

Also, less computation will be required, which reduces the aggregations of rounding/truncated errors.

1.4.1 Total Internal Reflection

A condition set within my code was if the magnitude of the parallel component of R squared ($(\|\vec{R}_{\parallel}\|)^2$ defined as `mag_refracted_parallel_sq`) was smaller than zero, the ray geometry would be handled as a reflection.

The derivation for this is as follows:

Using Pythagoras' Theorem it is known that:

$$1 = \sin^2 \phi + \cos^2 \phi$$

Where ϕ is defined as above.

Therefore:

$$1 = \|R_{\perp}\|^2 + \|R_{\parallel}\|^2$$

$\{\|R_{\parallel}\|^2, \|R_{\perp}\|^2\} \in [0, 1]$, where the elements of the vector coordinates are real

At the critical angle, θ_c :

$$n_1 \sin \theta_c = n_2$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

Let $\sin \theta_c \equiv C$. For $\theta > \theta_c$:

$$\frac{C n_1}{n_2} > 1$$

This will produce an error as $f(x) > 1$ is not in the range of $\sin(x)$, reflection within the current medium will occur.

Therefore: if $\|\vec{I}_{\perp} \frac{n_1}{n_2}\|^2 > 1$, then the ray will undergo total internal reflection (TIR). The magnitude is taken to represent the side length.

$$\vec{I}_{\perp} \frac{n_1}{n_2} \equiv \|\vec{R}_{\perp}\|^2 > 1$$

The reflected vector can be calculated simply as: $\vec{R}_{\text{reflect}} = \vec{I} - 2(\vec{I} \cdot \vec{N})\vec{N}$.

Reference: <https://physics.stackexchange.com/questions/340075/does-snells-law-still-apply-after-total-internal-reflection-has-occurred>

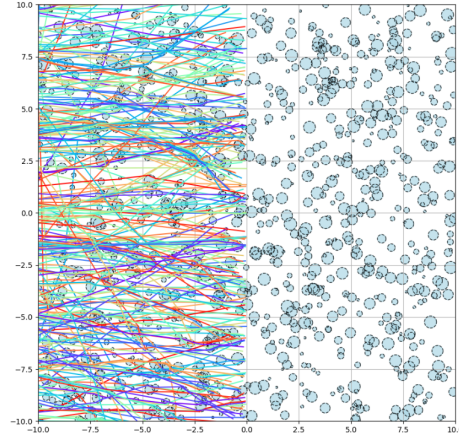


Figure 2: A screenshot of the ray tracing test that took place. The rays were able to handle all instances of refraction/reflection successfully

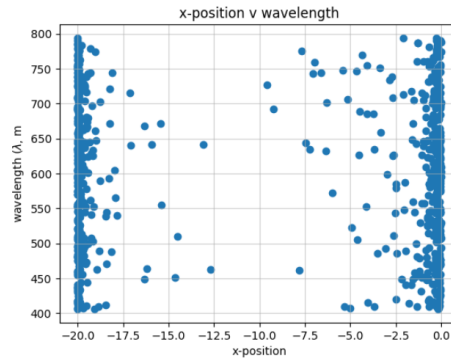


Figure 3: This is the initial plot of the wavelength against `x_position`. In the middle there was a faint correlation but did not quantify those

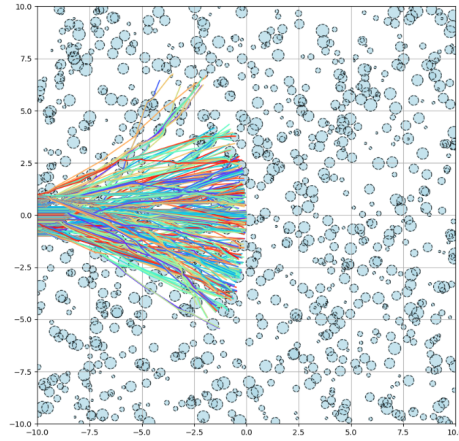


Figure 4: This plot shows a more concentrated beam of light dispersing with the raindrops. This more aptly represents Mie Scattering.

1.5 Data Analysis & Discussion

2 Conclusion

This project successfully demonstrates how geometric optics can approximate quantum scattering phenomena for educational purposes. While the model simplifies the complex electromagnetic interactions described by Mie theory, it captures the essential wavelength-dependent behavior that explains why shorter wavelengths scatter more effectively. The computational approach provides intuitive visualization of these effects and validates geometric optics as a useful educational tool for understanding atmospheric optics.